

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 10

DATE : 06-10-2019

Test Type : Minor

Test Pattern : AIIMS

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **180**.
5. **Each correct answer carry 1 mark. While one third mark will be deducted for each incorrect answer.**
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

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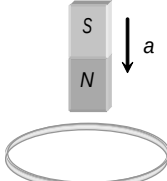
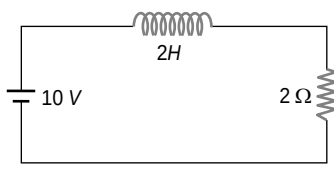
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Your Target is to secure Good Rank in AIIMS 2020

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

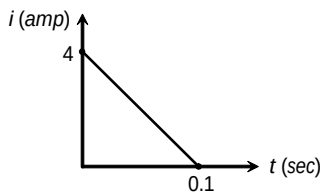
BEWARE OF NEGATIVE MARKING

- The magnetic flux linked with a coil is given by an equation ϕ (in *webers*) = $8t^2 + 3t + 5$. The induced e.m.f. in the coil at the fourth second will be
 (1) 16 *units* (2) 39 *units*
 (3) 67 *units* (4) 145 *units*
- A metallic ring is attached with the wall of a room. When the north pole of a magnet is brought near to it, the induced current in the ring will be (as observed from above)

 (1) First clockwise then anticlockwise
 (2) In clockwise direction
 (3) In anticlockwise direction
 (4) First anticlockwise then clockwise
- An ideal coil of 10 *henry* is joined in series with a resistance of 5 *ohm* and a battery of 5 *volt*. 2 second after joining, the current flowing in *ampere* in the circuit will be
 (1) e^{-1} (2) $(1 - e^{-1})$
 (3) $(1 - e)$ (4) e
- The SI unit of inductance, the henry, can be written as
 (1) *Weber/ampere*
 (2) *Volt-second/ampere*
 (3) *Joule/(ampere)²*
 (4) *Ohm-second*
- The self inductance of a solenoid of length L , area of cross-section A and having N turns is
 (1) $\frac{\mu_0 N^2 A}{L}$ (2) $\frac{\mu_0 N A}{L}$
 (3) $\mu_0 N^2 L A$ (4) $\mu_0 N A L$
- In the figure magnetic energy stored in the coil is

 (1) Zero
 (2) Infinite
 (3) 25 *joules*
 (4) None of the above
- Dynamo is a device for converting
 (1) Electrical energy into mechanical energy
 (2) Mechanical energy into electrical energy
 (3) Chemical energy into mechanical energy
 (4) Mechanical energy into chemical energy
- A transformer is employed to
 (1) Obtain a suitable dc voltage
 (2) Convert dc into ac
 (3) Obtain a suitable ac voltage
 (4) Convert ac into dc

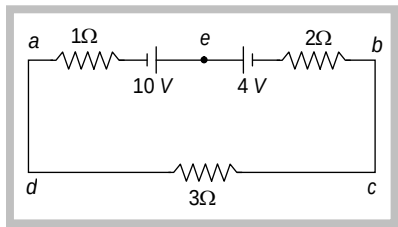
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9. The core of a transformer is laminated so that
- (1) Ratio of voltage in the primary and secondary may be increased
 - (2) Rusting of the core may be stopped
 - (3) Energy losses due to eddy currents may be reduced
 - (4) Change in flux is increased

10. Some magnetic flux is changed from a coil of resistance 10 ohm . As a result an induced current is developed in it, which varies with time as shown in figure. The magnitude of change in flux through the coil in *webers* is

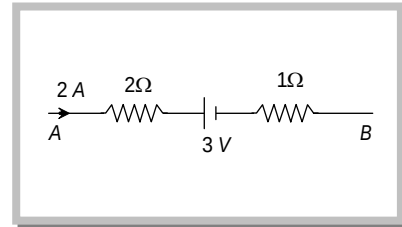


- (1) 2
 - (2) 4
 - (3) 6
 - (4) None of these
11. The magnitude and direction of the current in the circuit shown will be

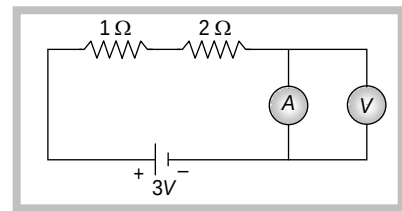


- (1) $\frac{7}{3} \text{ A}$ from a to b through e
- (2) $\frac{7}{3} \text{ A}$ from b and a through e
- (3) 1.0 A from b to a through e
- (4) 1.0 A from a to b through e

12. Figure represents a part of the closed circuit. The potential difference between points A and B ($V_A - V_B$) is



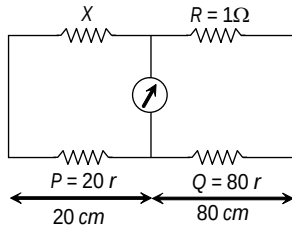
- (1) $+9 \text{ V}$
 - (2) -9 V
 - (3) $+3 \text{ V}$
 - (4) $+6 \text{ V}$
13. 100 mA current gives a full scale deflection in a galvanometer of resistance 2Ω . The resistance connected with the galvanometer to convert it into a voltmeter of 5 V range is
- (1) 98Ω
 - (2) 52Ω
 - (3) 80Ω
 - (4) 48Ω
14. In the circuit shown in figure, the voltmeter reading would be



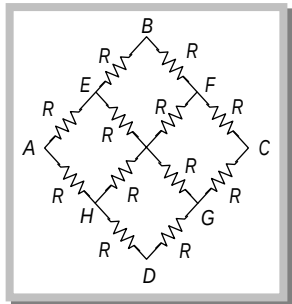
- (1) Zero
- (2) 0.5 volt
- (3) 1 volt
- (4) 2 volt

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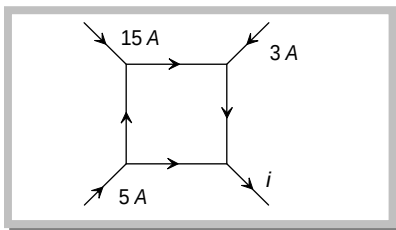
15. In meter bridge the balancing length from left and when standard resistance of $1\ \Omega$ is in right gap is found to be 20 cm . The value of unknown resistance is



- (1) $0.25\ \Omega$ (2) $0.4\ \Omega$
(3) $0.5\ \Omega$ (4) $4\ \Omega$
16. Following is the network of 12 identical resistances, equivalent resistance between point A and C is to be calculated.

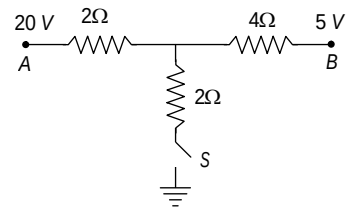


- (1) $\frac{R}{2}$ (2) $\frac{3R}{2}$ (3) $3R$ (4) $\frac{2R}{3}$
17. The figure shows a network of currents. The magnitude of current is shown here. The current i will be

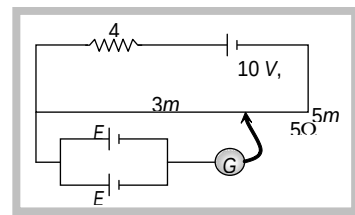


- (1) 3 A (2) 13 A
(3) 23 A (4) -3 A

18. As the switch S is closed in the circuit shown in figure, current passed through it is



- (1) 4.5 A (2) 6.0 A
(3) 3.0 A (4) Zero
19. A 2 volt battery, a $15\ \Omega$ resistor and a potentiometer of 100 cm length, all are connected in series. If the resistance of potentiometer wire is $5\ \Omega$, then the potential gradient of the potentiometer wire is
- (1) 0.005 V/cm
(2) 0.05 V/cm
(3) 0.02 V/cm
(4) 0.2 V/cm
20. A resistance of $4\ \Omega$ and a wire of length 5 metres and resistance $5\ \Omega$ are joined in series and connected to a cell of emf 10 V and internal resistance $1\ \Omega$. A parallel combination of two identical cells is balanced across 300 cm of the wire. The emf E of each cell is



- (1) 1.5 V (2) 3.0 V
(3) 0.67 V (4) 1.33 V

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21. A current of 1 mA is flowing through a copper wire. How many electrons will pass a given point in one second

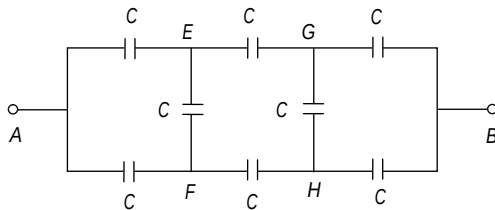
$[e = 1.6 \times 10^{-19} \text{ coulomb}]$

- (1) 6.25×10^{19}
(2) 6.25×10^{15}
(3) 6.25×10^{31}
(4) 6.25×10^9

22. In a step-down transformer, the transformation ratio is 0.1 , current in primary is 10 mA . The current in secondary is

- (1) 10 mA (2) 1 mA
(3) 1 mA (4) 0.1 A

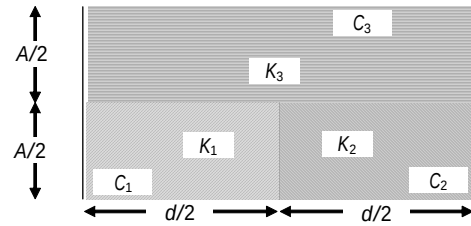
23. Find equivalent capacitance



24. The electric resistance of a certain wire of iron is R . If its length and radius are both doubled, then

- (1) The resistance will be halved and the specific resistance will be doubled
(2) The resistance and the specific resistance, will both remain unchanged
(3) The resistance will be doubled and the specific resistance will be halved
(4) The resistance will be halved and the specific resistance will remain unchanged

25. Equivalent dielectric constant of the combination



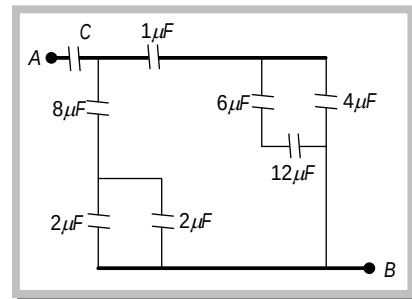
(1) $k_{eq} = \left(\frac{k_3}{2} + \frac{k_1 k_2}{k_1 + k_2} \right)$

(2) $k_{eq} = \left(\frac{k_3}{3} + \frac{k_1 k_2}{k_1 + k_2} \right)$

(3) $k_{eq} = \left(\frac{k_3}{2} + 2 \frac{k_1 k_2}{k_1 + k_2} \right)$

(4) $k_{eq} = \left(\frac{k_3}{4} + \frac{k_1 k_2}{k_1 + k_2} \right)$

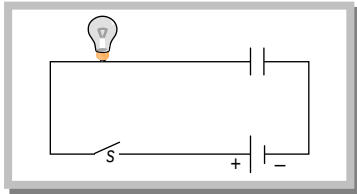
26. In the following figure the resultant capacitance between A and B is $1 \mu\text{F}$. The capacitance C is



- (1) $\frac{32}{11} \mu\text{F}$ (2) $\frac{11}{32} \mu\text{F}$
(3) $\frac{23}{32} \mu\text{F}$ (4) $\frac{32}{23} \mu\text{F}$

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27. A light bulb, a capacitor and a battery are connected together as shown here, with switch S initially open. When the switch S is closed, which one of the following is true

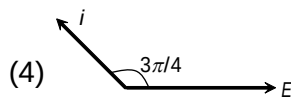
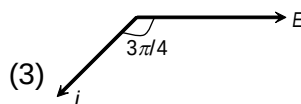
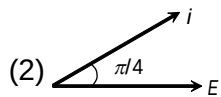
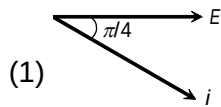


- (1) The bulb will light up for an instant when the capacitor starts charging
 (2) The bulb will light up when the capacitor is fully charged
 (3) The bulb will not light up at all
 (4) The bulb will light up and go off at regular intervals
28. Plate separation of a $15\mu\text{F}$ capacitor is 2 mm . A dielectric slab ($K = 2$) of thickness 1 mm is inserted between the plates. Then new capacitance is given by
 (1) $15\mu\text{F}$ (2) $20\mu\text{F}$
 (3) $30\mu\text{F}$ (4) $25\mu\text{F}$
29. The force between the plates of a parallel plate capacitor of capacitance C and distance of separation of the plates d with a potential difference V between the plates, is
 (1) $\frac{CV^2}{2d}$ (2) $\frac{C^2V^2}{2d^2}$
 (3) $\frac{C^2V^2}{d^2}$ (4) $\frac{V^2d}{C}$

30. Alternating current can not be measured by dc ammeter because
 (1) Average value of current for complete cycle is zero
 (2) ac changes direction
 (3) ac can not pass through dc ammeter
 (4) dc Ammeter will get damaged
31. An alternating voltage is represented as $E = 20\sin 300t$. The average value of voltage over one cycle will be
 (1) Zero (2) 10 volt
 (3) $20\sqrt{2}\text{ volt}$ (4) $\frac{20}{\sqrt{2}}\text{ volt}$
32. An ac supply gives 30 V r.m.s. which passes through a $10\ \Omega$ resistance. The power dissipated in it is
 (1) $90\sqrt{2}\text{ W}$ (2) 90 W
 (3) $45\sqrt{2}\text{ W}$ (4) 45 W
33. An alternating current is given by the equation $i = i_1 \cos \omega t + i_2 \sin \omega t$. The *r.m.s.* current is given by
 (1) $\frac{1}{\sqrt{2}}(i_1 + i_2)$ (2) $\frac{1}{\sqrt{2}}(i_1 + i_2)^2$
 (3) $\frac{1}{\sqrt{2}}(i_1^2 + i_2^2)^{1/2}$ (4) $\frac{1}{2}(i_1^2 + i_2^2)^{1/2}$
34. In general in an alternating current circuit
 (1) Average value of current is zero
 (2) Average value of square of current is zero
 (3) Average power dissipation is zero
 (4) Phase difference between voltage and current is zero

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35. In a certain circuit $E = 200 \cos(314 t)$ and $i = \sin(314 t + \pi/4)$. Their vector representation is



36. In a region of uniform magnetic induction $B = 10^{-2} \text{ Tesla}$, a circular coil of radius 30 cm and resistance $\pi^2 \text{ ohm}$ is rotated about an axis which is perpendicular to the direction of B and which forms a diameter of the coil. If the coil rotates at 200 rpm the amplitude of the alternating current induced in the coil is

- (1) $4\pi^2 \text{ mA}$ (2) 30 mA
(3) 6 mA (4) 200 mA

37. In a series LCR circuit, operated with an ac of angular frequency ω , the total impedance is

(1) $[R^2 + (L\omega - C\omega)^2]^{1/2}$

(2) $\left[R^2 + \left(L\omega - \frac{1}{C\omega} \right)^2 \right]^{1/2}$

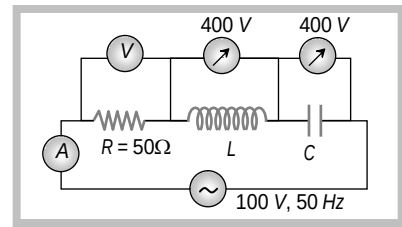
(3) $\left[R^2 + \left(L\omega - \frac{1}{C\omega} \right)^2 \right]^{-1/2}$

(4) $\left[(R\omega)^2 + \left(L\omega - \frac{1}{C\omega} \right)^2 \right]^{1/2}$

38. The band width of a series resonant circuit is 500 Hz and the resonant frequency is 5000 Hz . The quality factor of the circuit will be

- (1) 40 (2) 20
(3) 10 (4) 5

39. In the LCR series circuit the voltmeter and ammeter reading are



- (1) $V = 100 \text{ volt}, i = 2 \text{ amp}$
(2) $V = 100 \text{ volt}, i = 5 \text{ amp}$
(3) $V = 300 \text{ volt}, i = 2 \text{ amp}$
(4) $V = 300 \text{ V}, i = 1 \text{ amp}$

40. A copper disc of radius 0.1 m rotates about its centre with 10 revolutions per second in a uniform magnetic field of 0.1 Tesla . The emf induced across the radius of the disc is

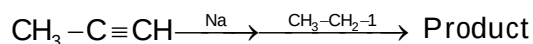
- (1) $\frac{\pi}{10} \text{ V}$ (2) $\frac{2\pi}{10} \text{ V}$
(3) $10 \pi \text{ mV}$ (4) $20 \pi \text{ mV}$

SPACE FOR ROUGH WORK

41. Which of the following halide is 2°?

- (1) Isopropyl chloride
- (2) Isobutyl chloride
- (3) n-propyl chloride
- (4) n-butyl chloride

42. What is the final product of reaction?



- (1) $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{CH}_3$
- (2) $\text{CH}_3 - \text{CH}_2 - \text{C} \equiv \text{C} - \text{CH}_2 - \text{CH}_3$
- (3) $\text{CH} \equiv \text{C} - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$
- (4) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_2 - \text{CH}_3$

43. Preparation of alkyl halides in laboratory is least preferred by

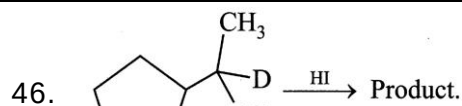
- (1) Treatment of alcohols
- (2) Addition of hydrogen halides to alkenes
- (3) Halide exchange
- (4) Direct halogenation of alkanes

44. Decreasing order of reactivity of HX in the reaction $\text{ROH} + \text{HX} \rightarrow \text{RX} + \text{H}_2\text{O}$

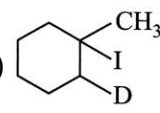
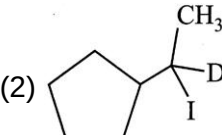
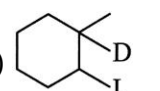
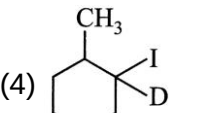
- (1) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
- (2) $\text{HBr} > \text{HCl} > \text{HI} > \text{HF}$
- (3) $\text{HCl} > \text{HBr} > \text{HI} > \text{HF}$
- (4) $\text{HF} > \text{HBr} > \text{HCl} > \text{HI}$

45. $\text{C}_2\text{H}_5\text{Cl} + \text{KCN} \rightarrow \text{X} \xrightarrow{\text{Hydrolysis}} \text{Y}$. 'X' and 'Y' are

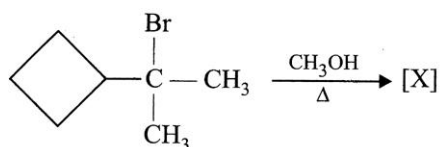
- (1) C_2H_6 and $\text{C}_2\text{H}_5\text{CN}$
- (2) $\text{C}_2\text{H}_5\text{CN}$ and C_2H_6
- (3) $\text{C}_2\text{H}_5\text{CN}$ and $\text{C}_2\text{H}_5\text{CH}_2\text{NH}_2$
- (4) $\text{C}_2\text{H}_5\text{CN}$ and $\text{C}_2\text{H}_5\text{COOH}$



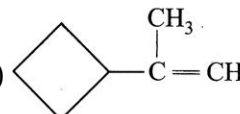
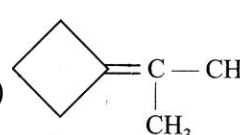
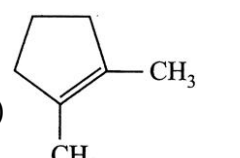
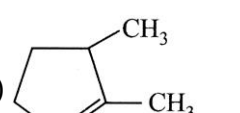
Identify the major product:

- (1) 
- (2) 
- (3) 
- (4) 

47. In the given reaction,



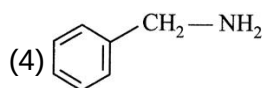
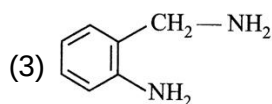
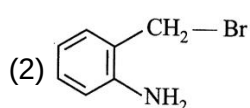
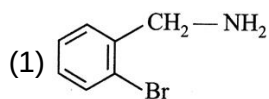
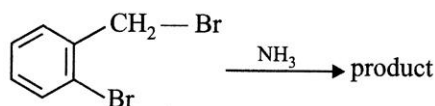
[X] as the major product among the elimination products is:

- (1) 
- (2) 
- (3) 
- (4) 

SPACE FOR ROUGH WORK

48. Iodoform heated with Ag powder to form
(1) Acetylene (2) Ethylene
(3) Methane (4) Ethane

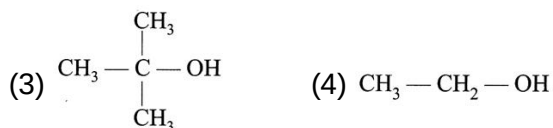
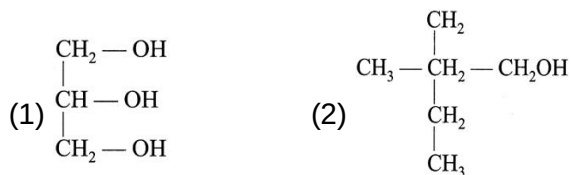
49. What is the major product obtained in the following reaction?



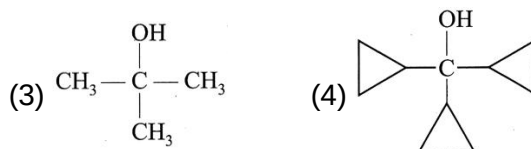
50. The product of the reaction
 $\text{Ph}_2\text{C}=\text{O} \xrightarrow[\text{H}_3\text{O}^+]{\text{LiAlD}_4}$ is

- (1) $\text{Ph}_2\text{CD}(\text{OH})$ (2) $\text{Ph}_2\text{CH}(\text{OD})$
(3) $\text{Ph}_2\text{CD}(\text{OD})$ (4) None of these

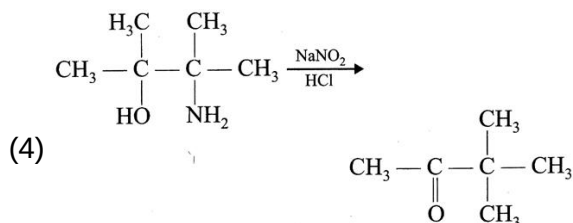
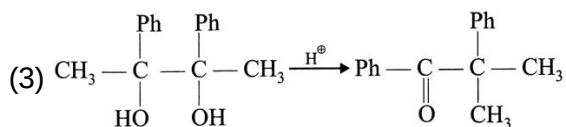
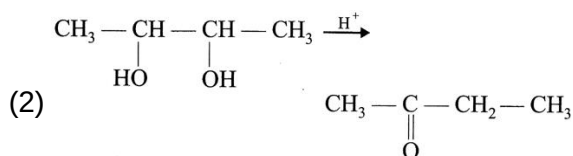
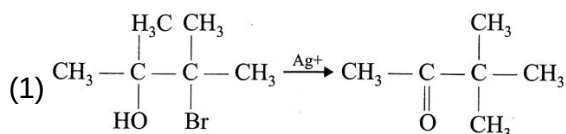
51. Which of the following is tertiary alcohol?



52. Which one of the following will be most reactive for $\text{S}_{\text{N}}1$ reaction?



53. Which of the following does not represent the correct product?



54. Which of the following compound give yellow precipitate with I_2 and NaOH ?

- (1) $\text{CH}_3\text{CH}_2\text{OH}$
(2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
(3) $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$
(4) CH_3OH

SPACE FOR ROUGH WORK

55. Phenyl magnesium bromide reacts with methanol to give:
- A mixture of anisole and Mg(OH)Br
 - A mixture of benzene and Mg(OMe)Br
 - A mixture of toluene and Mg(OH)Br
 - A mixture of phenol and Mg(Me)Br
56. The reaction of $\text{CH}_3\text{CH}=\text{CH}-\text{C}_6\text{H}_4-\text{OH}$ with HBr gives
- $\text{CH}_3\text{CHBrCH}_2-\text{C}_6\text{H}_4-\text{OH}$
 - $\text{CH}_3\text{CH}_2\text{CHBr}-\text{C}_6\text{H}_4-\text{OH}$
 - $\text{CH}_3\text{CHBrCH}_2-\text{C}_6\text{H}_4-\text{Br}$
 - $\text{CH}_3\text{CH}_2\text{CHBr}-\text{C}_6\text{H}_4-\text{Br}$
57. Phenol can be distinguished from aliphatic alcohol with
- Tollens reagent
 - Schiff's base
 - FeCl_3
 - HCl
58. Phenol is less acidic than
- Ethanol
 - o-nitrophenol
 - Methanol
 - p-methylphenol
59. One mole of phenol reacts with bromine to form tribromophenol. How much bromine is used?
- 1.5 mol
 - 4.5 mol
 - 3 mol
 - 6 mol
60. The Williamson ether synthesis follows the following mechanism-
- Nucleophilic substitution
 - Nucleophilic addition
 - Electrophilic addition
 - Electrophilic substitution
61. In the following reaction 'A' is
- $$\text{C}_2\text{H}_5\text{MgBr} + \text{H}_2\text{C}-\text{CH}_2 \xrightarrow{\text{H}_2\text{O}} \text{A}$$
- (The epoxide is shown as a three-membered ring with an oxygen atom.)
- $\text{C}_2\text{H}_5\text{CH}_2\text{CHO}$
 - $\text{C}_2\text{H}_5\text{CH}_2\text{OH}$
 - $\text{C}_2\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$
 - $\text{C}_2\text{H}_5\text{CHO}$
62. In the reaction $\text{C}_6\text{H}_5-\text{OCH}_3 \xrightarrow{\text{HBr}}$ the products are
- $\text{C}_6\text{H}_5-\text{OCH}_3$ and H_2
 - $\text{C}_6\text{H}_5-\text{Br}$ and CH_3Br
 - $\text{C}_6\text{H}_5-\text{Br}$ and CH_3OH
 - $\text{C}_6\text{H}_5-\text{OH}$ and CH_3Br

SPACE FOR ROUGH WORK

63. Which of the following will not form a yellow precipitate on heating with an alkaline solution of iodine

- (1) CH_3OH
- (2) $\text{CH}_3\text{CH}_2\text{OH}$
- (3) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
- (4) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$

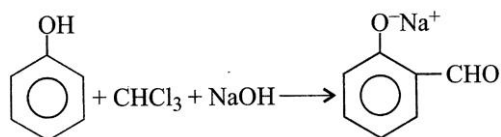
64. Which of the following compounds has the most acidic nature?

- (1)
- (2)
- (3)
- (4)

65. Which of the following will not be soluble in NaHCO_3 ?

- (1) 2,4,6-Trinitrophenol
- (2) Benzoic acid
- (3) o-Nitrophenol
- (4) Benzene sulphonic acid

66. In the reaction



The electrophile involved is

- (1) dichloromethyl cation (CHCl_2^+)
- (2) formyl cation (CHO^+)
- (3) dichloromethyl anion (CHCl_2^-)
- (4) dichlorocarbene ($:\text{CCl}_2$)

67. Compound which gives acetone on ozonolysis

- (1) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$
- (2) $(\text{CH}_3)_2\text{C} = \text{C}(\text{CH}_3)_2$
- (3) $\text{C}_6\text{H}_5\text{CH} = \text{CH}_2$
- (4) $\text{CH}_3\text{CH} = \text{CH}_2$

68. Catalyst SnCl_2/HCl is used in

- (1) Stephen's reduction
- (2) Cannizzaro reaction
- (3) Clemmensen's reduction
- (4) Rosenmund's reduction

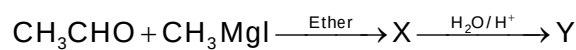
69. $\text{CH}_3 - \text{CH}_2 - \text{C} \equiv \text{CH} \xrightarrow[\text{H}_2\text{SO}_4]{\text{HgSO}_4} \text{A}$, the compound A is

- (1) $\text{CH}_3 - \text{CH}_2 - \text{CO} - \text{CH}_3$
- (2) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CHO}$
- (3) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{COOH}$
- (4) None of these

70. Which one of the following will most readily be dehydrated in acidic condition?

- (1)
- (2)
- (3)
- (4)

71. Identify the product Y in the sequence



- (1) CH_3OH
- (2) $\text{CH}_3\text{CH}_2\text{OH}$
- (3) $(\text{CH}_3)_2\text{CHOH}$
- (4) $(\text{CH}_3)_3\text{COH}$

SPACE FOR ROUGH WORK

72. The reagent that gives an orange coloured precipitate with acetaldehyde is
 (1) NH_2OH
 (2) NaHSO_3
 (3) Iodine
 (4) 2,4-DNP
73. $\text{C}_2\text{H}_5\text{CHO}$ and $(\text{CH}_3)_2\text{CO}$ can be distinguished by testing with
 (1) Phenyl hydrazine
 (2) Hydroxylamine
 (3) Fehling solution
 (4) Sodium bisulphite
74. The Clemmensen reduction of acetone yields
 (1) Ethanol
 (2) Ethanal
 (3) Propane
 (4) Propanol
75. Which of the following organic compounds exhibits positive Fehling test as well as iodoform test?
 (1) Methanal
 (2) Ethanol
 (3) Propanone
 (4) Ethanal
76. The alkaline CuSO_4 containing sodium potassium tartrate does not react with
 (1) CH_3CHO
 (2) $\text{C}_2\text{H}_5\text{CHO}$
 (3) $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$
 (4) $\text{C}_6\text{H}_5\text{CHO}$
77. The number of aldol reaction(s) that occurs in the given transformation is

$$\text{CH}_3\text{CHO} + 4\text{HCHO} \xrightarrow{\text{con. aq. NaOH}} \begin{array}{c} \text{OH} \quad \text{OH} \\ | \quad | \\ \text{HO}-\text{C}-\text{C}-\text{OH} \\ | \quad | \\ \text{HO} \quad \text{OH} \end{array}$$

 (1) 1 (2) 2
 (3) 3 (4) 4
78. In the Cannizzaro reaction given below,

$$2\text{PhCHO} \xrightarrow{^-\text{OH}} \text{PhCH}_2\text{OH} + \text{PhCO}_2^-$$

 the slowest step is
 (1) the attack of ^-OH at the carbonyl group,
 (2) the transfer of hydride to the carbonyl group,
 (3) the abstraction of proton from the carboxylic acid,
 (4) the deprotonation of PhCH_2OH .
79. Which one of the following orders of acid strength is correct?
 (1) $\text{RCOOH} > \text{HC} \equiv \text{CH} > \text{HOH} > \text{ROH}$
 (2) $\text{RCOOH} > \text{ROH} > \text{HOH} > \text{HC} \equiv \text{CH}$
 (3) $\text{RCOOH} > \text{HOH} > \text{ROH} > \text{HC} \equiv \text{CH}$
 (4) $\text{RCOOH} > \text{HOH} > \text{HC} \equiv \text{CH} > \text{ROH}$
80. Nitration of benzoic acid gives
 (1) 3-nitrobenzoic acid
 (2) 2-nitrobenzoic acid
 (3) 2, 3-dinitrobenzoic acid
 (4) 2, 4-dinitrobenzoic acid

SPACE FOR ROUGH WORK

81. If a plant with genotype Tt is self-pollinated. What is the probability of pollen with T genotype fused with egg which contain t genotype.
- (1) 25% (2) 50%
(3) 75% (4) 100%
82. Plant with genotype $AaBbCc$ crossed with $AABBCC$ then find out probability of plant with genotype $AABbCC$:-
- (1) $\frac{2}{64}$ (2) $\frac{4}{64}$
(3) $\frac{1}{8}$ (4) $\frac{3}{8}$
83. Find out phenotypic ratio in F_2 generation of Pea plant when considered size of starch grain and shape of seed as a character respectively

(1)	Phenotypic ratio = 1:2:1 Phenotypic ratio = 3:1	(For starch grain size) (for seed shape)
(2)	Phenotypic ratio = 3:1 Phenotypic ratio = 3:1	(For starch grain size) (for seed shape)
(3)	Phenotypic ratio = 1:2:1 Phenotypic ratio = 1:2:1	(For starch grain size) (for seed shape)
(4)	Phenotypic ratio = 3:1 Phenotypic ratio = 1:2:1	(For starch grain size) (for seed shape)

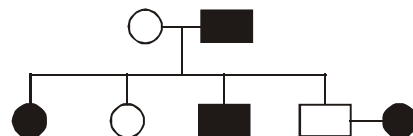
84. The value of X/A is given in column A and sex of the fruit fly is written in column B. Match the following :-

Column-A		Column-B	
1	1	a	Normal Male
2	1.5	b	Super male sterile
3	0.67	c	Inter sex
4	0.33	d	Super female sterile
5	0.5	e	Normal female

Options :-

- (1) (1-a), (2-d), (3-c), (4-e), (5-b)
(2) (1-e), (2-d), (3-c), (4-b), (5-a)
(3) (1-b), (2-a), (3-c), (4-d), (5-e)
(4) (1-e), (2-c), (3-d), (4-a), (5-b)

85. In an experiment individual homozygous for ab genes were crossed with wild type ($++$). The F_1 hybrid thus produced was test crossed and progenies produced in following ratio.
- $++548, ab340, +a44, +b68$
- Calculate the distance between b and a genes.
- (1) 46 m.u.
(2) 22.4 m.u.
(3) 44 m.u.
(4) 11.2 m.u.
86. Ratio of coupling is :-
- (1) 1 : 1 : 1 : 1
(2) 7 : 1 : 1 : 7
(3) 1 : 7 : 7 : 1
(4) 9 : 3 : 3 : 1
87. Split genes include
- (1) Exons
(2) Introns
(3) Pseudoalleles
(4) Both (1) and (2)
88. Find out the genotype of father and mother in the given pedigree chart



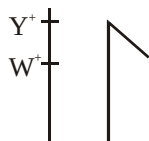
	Mother	Father
(1)	AA	Aa
(2)	Aa	Aa
(3)	AA	aa
(4)	aa	Aa

89. Which of the following represent normal female in *Drosophila*?
- (1) $2A + XX$
(2) $2A + XY$
(3) $2A + XO$
(4) $3A + XO$

90. In a random mating population frequency of recessive allele is 0.5. What is the frequency of dominant phenotypes in population :-

- (1) 0.25 (2) 0.50
(3) 0.75 (4) 0.20

91. How many type of gametes will be produced by ♂ drosophila having following arrangement of two genes (Y^+ and W^+) on X-chromosomes :-



- (1) 2 (2) 4
(3) 1 (4) 8

92. Sequencing the whole set of genome that contained all the coding and non-coding sequence and later assigning different region in the sequence with functions are referred to as :-

- (1) Expressed Tags
(2) Sequence Annotation
(3) SETs
(4) SNP's

93. In operon model, RNA polymerase binds to

- (1) Structural gene
(2) Promoter gene
(3) Regulator
(4) Operator gene

94. Largest gene in humans is :-

- (1) Dystrophin
(2) Insulin gene
(3) Oncogene
(4) x-globin of haemoglobin

95. Select the correct sequence of steps in DNA finger printing involving Southern blot hybridisation using radiolabelled VNTR as probe.

I. Hybridisation using labelled VNTR probe.

II. Isolation of DNA.

III. Transferring (blotting) of separated DNA fragments to synthetic membranes, such as nitrocellulose or nylon.

IV. Detection of hybridisation DNA fragments by autoradiography.

V. Separation of DNA fragments by electrophoresis.

VI. Digestion of DNA by restriction endonucleases.

(1) I, V, VI, II, III and IV

(2) II, VI, V, III, I and IV

(3) V, I, VI, III, IV and II

(4) II, I, V, VI, IV, III

96. Human genome project was started in

- (1) 1989 (2) 1990
(3) 1992 (4) 1995

97. Nuclear DNA sends information for protein synthesis through.

- (1) tRNA
(2) mRNA
(3) rRNA
(4) All the above

98. In *Drosophila melanogaster*, the genes white and yellow showsa..... recombination and genes white and miniature wing showsb..... linkage

- (1) $a \rightarrow 98.7\%$, $b \rightarrow 37.2\%$
(2) $a \rightarrow 98.7\%$, $b \rightarrow 62.8\%$
(3) $a \rightarrow 1.3\%$, $b \rightarrow 37.2\%$
(4) $a \rightarrow 1.3\%$, $b \rightarrow 62.8\%$

99. A tall pea plant round seeds (TTRR) is crossed with a dwarf wrinkle seeded plant (ttrr). F_1 has tall plants with rounded seeds. What is the proportion of dwarf plants with wrinkled seeds in F_2 generation?

- (1) Zero (2) $1/2$
(3) $1/4$ (4) $1/16$

100. In quantitative inheritance F_2 ratio obtained in a dihybrid cross is :-

- (1) $1 : 4 : 6 : 4 : 1$ (2) $15 : 1$
(3) $12 : 3 : 1$ (4) $9 : 7$

101. A female suffering from patau syndrome will have how many barr bodies ?
- (1) 2 (2) 1
(3) 0 (4) 3
102. Mark the incorrect matching :-
- (1) Down syndrome - Aneuploidy
(2) Turner syndrome - 45 with XO
(3) Cri-du-chat - Trisomy in chromosome 5th
(4) Klinefelter syndrome - 44 + XXY
103. Pannus Granules are formed in ?
- (1) Masthenia gravis
(2) IDDM
(3) Rheumatoid arthritis
(4) Pernicious anemia
104. Salk polio vaccine is _____
- (1) Live
(2) Killed
(3) Toxoid
(4) Recombinant
105. Intestinal perforation and death may occur in severe cases of which disease ?
- (1) Pneumonia
(2) Typhoid
(3) Filariasis
(4) Malaria
106. Malignant malaria is caused by ?
- (1) *Plasmodium vivax*
(2) *Plasmodium falciparum*
(3) *Plasmodium malariae*
(4) Retro virus
107. Good Humor hypothesis of health was given by Hippocrates. Select the correct statements:-
- (a) Black bile persons belonged to hot personality
(b) Persons which have black bile have fevers
(c) Persons with yellow bile belong to hot personality
(d) William Harvey disproved good humor hypothesis
(e) Harvey disproved this theory with help of thermometer
- (1) a, b
(2) a, b, c
(3) a, b, d, e
(4) a, b, c, d, e
108. True about plasma cells?
- (1) Are derived from T-cells
(2) Develop into B-cells
(3) Secrete large amount of Beta interferones.
(4) Specifically forms antibodies against particular pathogen.
109. Injecting microbes deliberately into host body to induce immunity is the part of?
- (1) Active immunisation
(2) Passive immunisation
(3) Non-specific immunity
(4) Innate immunity
110. Which of the following are included in prophylaxis?
- a. Antiserum b. Toxoid
c. Vaccines d. Antivenom
e. Triple antigen (DPT,MMR)
- Choose the correct option from below
- (1) a, b, c, d, e
(2) b, c, e
(3) a, c, d, e
(4) a, b, c, d

111. After immunofluorescence technique done on stomach sections, Antibodies found in parietal cells are marker for ?
- (1) Thyrotoxicosis
 - (2) Pernicious Anaemia
 - (3) Iron deficiency Anaemia
 - (4) Thalassemia
112. Which of the following generation of vaccine prepared by recombinant DNA technology?
- (1) Ist generation
 - (2) IInd generation
 - (3) IIIrd generation
 - (4) IVth generation
113. Which of the following two statements correct for kidney transplant ?
- (a) Immuno suppressant drugs used for long time inspite kidney properly transplanted
 - (b) Graft rejection is due to cell mediated immunity
 - (c) B-cells responsible for graft rejection
 - (d) Acceptance and rejection of graft depends on specific interferones.
- (1) (a) and (b)
 - (2) (b) and (d)
 - (3) (c) and (d)
 - (4) (a) and (d)
114. In case of IDDM, why insulin not given Intravenously ?
- (1) It is not effective and glucose uptake not happens
 - (2) Chances of Insulin shock
 - (3) Insulin more effectively works in other routes
 - (4) Intradermal is the preferred route
115. Mark the incorrect matching :-
- (1) Class-I MHC molecule = CD-8 receptor
 - (2) Class-II MHC molecule = CD-4 receptor
 - (3) MHC-I = Endogenous proteins
 - (4) MHC-I = Made primarily by APCs
116. _____ constitutes about 50 percent of the lymphoid tissue in human body :-
- (1) Primary lymphoid organs
 - (2) Spleen
 - (3) M.A.L.T.
 - (4) Bone marrow
117. A patient consumed contaminated food few days before, now he is suffering from sustained high fever, headache, weakness, loss of appetite. What can be the probable disease?
- (1) Typhoid
 - (2) Pneumonia
 - (3) Diphtheria
 - (4) Tuberculosis
118. The rupture of RBC is associated with the release of a toxin
- (1) Hirudin
 - (2) Anophelin
 - (3) Heparin
 - (4) Haemozoin
119. Which type of bond is present in between light and heavy chain of an antibody?
- (1) Ionic bond
 - (2) Peptide bond
 - (3) Dioxide bond
 - (4) Disulphide bond
120. Acidity in human stomach is an example of
- (1) Cellular barrier
 - (2) Antibody mediated immunity
 - (3) Physiological barrier
 - (4) Cell mediated immunity

Direction for Assertion & Reason questions. These questions consist of two statements each, printed as assertion and reason. while answering these questions you are required to choose any one of the following four responses.

- (1) If both Assertion & Reason are true & the reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are true but reason is not a correct explanation of the Assertion.
- (3) If Assertion is true but the Reason is false
- (4) If both Assertion & Reason are false

121. *Assertion* : Only a change in magnetic flux will maintain an induced current the coil.

Reason : The presence of large magnetic flux through a coil maintains a current in the coil if the circuit is continuous.

122. *Assertion* : Faraday established induced e.m.f. experimentally.

Reason : Magnetic flux can produce induced e.m.f.

123. *Assertion* : The induced e.m.f. and current will be different in two identical loops of copper and aluminium, when rotated with same speed in the same magnetic field.

Reason : Induced EMF and current both depend only on rate of change of magnetic flux.

124. *Assertion* : Inductance coil are made of copper.

Reason : Induced current is more in wire having less resistance.

125. *Assertion* : When two coils are wound on each other, the mutual induction between the coils is maximum.

Reason : Mutual induction does not depend on the orientation of the coils.

126. *Assertion* : Acceleration of a magnet falling through a long solenoid decreases.

Reason : The induced current produced in a circuit always flow in such direction that it opposes the change or the cause the produced it.

127. *Assertion* : The dc and ac both can be measured by a hot wire instrument.

Reason : The hot wire instrument is based on the principle of magnetic effect of current.

128. *Assertion* : Induced e.m.f., opposes always the change in magnetic flux responsible for its production.

Reason : Lenz's law violates the principle of conservation of energy.

129. *Assertion* : The induced emf in a conducting loop of wire will be non zero when it rotates in a uniform magnetic field.

Reason : The emf is induced due to change in magnetic flux.

130. *Assertion* : An induced emf is generated when magnet is withdrawn from the solenoid.

Reason : The relative motion between magnet and solenoid induces emf.

131. *Assertion* : A transformer cannot work on dc supply.

Reason : dc changes neither in magnitude nor in direction.

132. *Assertion* : Soft iron is used as a core of transformer.

Reason : Area of hysteresis is loop for soft iron is small.

133. *Assertion* : An ac generator is based on the phenomenon of self-induction.

Reason : In single coil, we consider mutual-induction also.

134. *Assertion* : In series LCR circuit resonance can take place.

Reason : Resonance takes place if inductance and capacitive reactances are equal and opposite.

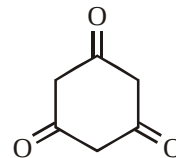
135. *Assertion* : The alternating current lags behind the e.m.f. by a phase angle of $\pi/2$, when ac flows through an inductor.

Reason : The inductive reactance increases as the frequency of ac source decreases.

136. *Assertion* : Capacitor serves as a block for dc and offers an easy path to ac.
Reason : Capacitive reactance is inversely proportional to frequency.
137. *Assertion* : When capacitive reactance is smaller than the inductive reactance in LCR current, e.m.f. leads the current .
Reason : The phase angle is the angle between the alternating e.m.f. and alternating current of the circuit.
138. *Assertion* : Chock coil is preferred over a resistor to adjust current in an ac circuit.
Reason : Power factor for inductance is zero.
139. *Assertion* : If the frequency of alternating current in an ac circuit consisting of an inductance coil is increased then current gets decreased.
Reason : The current is inversely proportional to frequency of alternating current in the inductor coil.
140. *Assertion* : An alternating current does show magnetic effect.
Reason : Alternating current varies with time.
141. *Assertion* : RI is more reactive than RCl towards S_N reaction.
Reason : The rate of reaction for S_N1 or S_N2 mechanism is $RCl > RI$.
142. *Assertion* : $CH_3Br + AgCN \rightarrow CH_3NC + AgBr$.
Reason : CN^- is an ambident ion.
143. *Assertion* : S_N2 reaction proceed with inversion of configuration.
Reason : S_N2 reactions occur in one step.
144. *Assertion* : Aniline is more basic than pyrrole.
Reason : Aniline and pyrrole both are aromatic.
145. *Assertion* : Nucleophilic substitution on an optically active alkyl halide gives a mixture of enantiomers.
Reason : The reaction occurs according to S_N2 mechanism.

146. *Assertion* : Butene-1 on reaction with HBr in the presence of a peroxide produces 1-bromobutane.

Reason : It involves formation of a primary radical.



147. *Assertion* : , will show tautomerism.

Reason : It contains α -hydrogen atom near keto group.

148. *Assertion* : Primary and secondary alcohols can be distinguished by Victor-Meyer's test.

Reason : Primary alcohols form nitrolic acid which dissolve in NaOH to form blood red colouration but secondary alcohols form pseudonitrates which give blue colouration with NaOH.

149. *Assertion* : Cyclohexanone exhibits keto-enol tautomerism.

Reason : Cyclohexanone, has molecular formula $C_6H_{10}O$.

150. *Assertion* : Esters which contain α -hydrogens undergo Claisen condensation.

Reason : $LiAlH_4$ reduction of esters gives acids.

151. *Assertion* : The acetate ion is resonance stabilized.

Reason : Acetate ion is more basic than the methoxide

152. *Assertion* : Aromatic aldehydes can be distinguished from aliphatic aldehydes by Fehling's solution test.

Reason : Fehling's solution is an alkaline solution of $CuSO_4$ containing Rochelle salt.

153. *Assertion* : The addition of ammonia derivative to a carbonyl compound is carried out in weakly acidic medium.

Reason : In weakly acidic medium attacking nucleophile is also protonated.

- | | |
|---|--|
| <p>154. <i>Assertion</i> : CCl_3CHO forms an isolable crystalline hydrate.
<i>Reason</i> : Electron withdrawing chlorine atoms stabilise hydrate by intramolecular H-bonding.</p> <p>155. <i>Assertion</i> : 2, 2-Dimethylpropanal undergoes Cannizzaro reaction with conc. NaOH.
<i>Reason</i> : Cannizzaro reaction is a disproportionation reaction.</p> <p>156. <i>Assertion</i> : Acetic acid on reaction with hydrazoic acid (N_3H) in the presence of H_2SO_4 followed by heating and hydrolysis gives acetamide.
<i>Reason</i> : Methyl cyanate ($\text{Me}-\text{CNO}$) is formed as an intermediate compound.</p> <p>157. <i>Assertion</i> : Amines are basic in nature
<i>Reason</i> : Presence of lone pair of electrons on nitrogen atom.</p> <p>158. <i>Assertion</i> : Benzene diazonium salt on boiling with water forms phenol.
<i>Reason</i> : Benzene diazonium salts are more stable than aliphatic diazonium salts.</p> <p>159. <i>Assertion</i> : Phenol reacts with bromine water to form 2, 4, 6-tribromophenol.
<i>Reason</i> : Phenol is resonance stabilized.</p> <p>160. <i>Assertion</i> : Aniline does not undergo Friedel-Crafts reaction.
<i>Reason</i> : Friedel-Crafts is an electrophilic substitution reaction.</p> <p>161. <i>Assertion</i> : Starch synthesis in pea seeds is controlled by one gene.
<i>Reason</i> : This gene has two alleles (B and b).</p> <p>162. <i>Assertion</i> : Dominance is not an autonomous feature of a gene or the product that it has information for.
<i>Reason</i> : Dominance depends much on the gene product and the production of a particular phenotype from this product.</p> | <p>163. <i>Assertion</i> : The individuals affected by Klinefelter's or Turner's syndrome.
<i>Reason</i> : Klinefelter's and Turner's syndrome are due to chromosomal aberration.</p> <p>164. <i>Assertion</i> : $\text{Hb}^S \text{Hb}^S$ is homozygous condition of sickle cell anaemia.
<i>Reason</i> : It occurs due to substitution of glutamic acid by valine at sixth position in β-chain of haemoglobin.</p> <p>165. <i>Assertion</i> : If the sequence of bases in one strand is known then the sequence in other strand can be predicted.
<i>Reason</i> : Both the strands of DNA are complementary of each other.</p> <p>166. <i>Assertion</i> : In transforming principle, when <i>Streptococcus pneumoniae</i> (<i>Pneumococcus</i>) bacteria are grown on a culture plate, some produce smooth shiny colonies (S) while other produce rough colonies (R)
<i>Reason</i> : S strain bacteria have a mucous (polysaccharide) coat, while R strain does not.</p> <p>167. <i>Assertion</i> : DNA is preferred over RNA for storage of genetic information.
<i>Reason</i> : DNA is more stable than RNA.</p> <p>168. <i>Assertion</i> : RNA being a catalyst was reactive and hence unstable.
<i>Reason</i> : DNA has evolved from RNA with chemical modification that make it more stable.</p> <p>169. <i>Assertion</i> : Split genes are found in eukaryotes.
<i>Reason</i> : Introns or intervening sequences are found in prokaryotes.</p> <p>170. <i>Assertion</i> : The transcription and translation can be coupled in bacteria.
<i>Reason</i> : In bacteria transcription and translation take place in the same compartment and also the mRNA does not require any processing to become active.</p> |
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171. <i>Assertion</i> : Health, for a long time, was considered as a state of body and mind where there was a balance of certain humors. <i>Reason</i> : This is what early Greeks like Hippocrates as well as Indian Ayurveda system of medicine asserted. 172. <i>Assertion</i> : Previously, it was thought that persons with 'blackbile' belonged to hot personality and would have fevers. <i>Reason</i> : The demonstration of normal body temperature in persons with blackbile using thermometer approved the 'good humor' hypothesis. 173. <i>Assertion</i> : Health is affected by genetic disorders, infections and life style <i>Reason</i> : Mind and mental state affect our health. 174. <i>Assertion</i> : Health is defined as a state of complete physical, mental and social well-being. <i>Reason</i> : Health is only defined as a state of physical fitness/absence of disease. 175. <i>Assertion</i> : In severe cases, the lips and finger nails may turn gray to bluish in colour in common cold. <i>Reason</i> : Dysentery, plague, diphtheria and typhoid are viral diseases.	176. <i>Assertion</i> : Innate immunity is specific type of defence that present at the time of birth. <i>Reason</i> : Acquired immunity is non-specific type of defence which is characterised by memory. 177. <i>Assertion</i> : PMNL are type of lymphocytes and natural killer cells <i>Reason</i> : Macrophages in the tissues are physiological barriers. 178. <i>Assertion</i> : Each antibody is represented as H_2L_2. <i>Reason</i> : Each antibody molecule has four nucleotide chains, two small called light chains and two longer called heavy chains. 179. <i>Assertion</i> : Hepatitis-B vaccine produced from yeast. <i>Reason</i> : Recombinant DNA technology has allowed the production of antigenic polypeptides of pathogen in bacteria or yeast. 180. <i>Assertion</i> : Symptoms of the allergic reactions include sneezing, watery eyes, running nose and difficulty in breathing. <i>Reason</i> : Allergy is due to chemicals like anti-histamine, adrenalin and steroids.
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